

OAC Thunder common bean

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Michaels, T. E. and Smith, T. H. 1999. **OAC Thunder common bean**. Can. J. Plant Sci. **79**: 101–102. OAC Thunder is an indeterminate bush white bean (*Phaseolus vulgaris* L.) cultivar intended for use in areas with greater than 2600 crop heat units. It has excellent yield potential in either wide or narrow row production and is resistant to races 1 and 15 of bean common mosaic virus. Seed has acceptable cooking and canning quality.

Key words: *Phaseolus vulgaris* L., white bean, common bean, cultivar description

Michaels, T. E. et Smith, T. H. 1999. **Le haricot OAC Thunder**. Can. J. Plant Sci. **79**: 101–102. OAC Thunder est un cultivar nain de haricot blanc (*Phaseolus vulgaris* L.) à type de croissance indéfinie destiné aux régions recevant plus de 2 600 unités thermiques de croissance. Il affiche une excellente productivité en lignes à écartement large ou à écartement étroit et il est résistant aux races 1 et 15 du virus de la mosaïque commune du haricot. Le grain convient pour la cuisson et pour la mise en conserve.

Mots clés: *Phaseolus vulgaris* L., haricot blanc, haricot commun, description de cultivar

The white seeded common bean (*Phaseolus vulgaris* L.) cultivar OAC Thunder was developed by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario. It is intended for production in areas of Ontario with more than 2600 crop heat units. OAC Thunder was issued Registration No. 4491 on 3 March 1997 by the Registrar, Variety Registration Office, Plant Products Division, Agriculture and Agri-Food Canada, Nepean, Ontario.

Pedigree and Breeding Methods

OAC Thunder, tested as OAC 93-1, was derived from the cross Ex Rico subline 1351/W84-129 made in 1986 in a growth room. Ex Rico subline 1351 is a breeding line with pedigree Ex Rico 23*5//Ex Rico 23/Narda. W84-129 is an F₃ single plant selection made in 1985 from a population with the pedigree T81-01/Domino//Ex Rico Are/Domino/3/NZ Upright/Sacramento Light Red Kidney. F₁ plants were grown in the field in 1986 at the Elora Research Station and all seeds were bulked. The F₂ to F₄ generations were advanced using the single pod descent (pod bulk) method. The F₂ and F₄ generations were grown utilising a winter nursery in New Zealand in the winters of 1986–1987 and 1987–1988. The F₃ generation was grown at the Elora Research Station in the summer of 1987. Single plant selections for early maturity, upright plant architecture, white bean seed type and high pod number were made from space planted F₅ bulk plots in the field at Elora, Ontario in 1988. F₆ progeny rows were grown and selected for the same agronomic traits at the Elora Research Station in 1989. F₇ and F₈ generations were grown in 1990 and 1991 at Woodstock and Elora, Ontario in yield trials, where selection was for the

same agronomic traits plus seed yield under conventional and narrow row production with direct combining. The line OAC 93-1 was identified in these trials and entered in the Ontario Pulse Committee's White Bean Cultivar Trials in 1993 and 1994. One hundred plants were grown in the winter of 1994–1995 in a growth room from residual F₆ seed, and the progeny of those plants were sent to Twin Falls, Idaho, USA, for breeder seed production in 1995 and 1996. Breeder seed plots were rogued for uniformity and trueness to type, and seed from the remaining plants were bulked to produce breeder seed.

OAC Thunder was tested for resistance to bean common mosaic virus by artificial inoculation in growth rooms at the University of Guelph. Cooking quality was assessed at Agriculture and Agri-Food Canada Greenhouse and Processing Crops Research Centre, Harrow, Ontario by a panel for appearance, flavour and texture. A Hunter Labscan was used to determine cooked bean colour as L (white to black), aL (red to green) and bL (yellow to blue). Canned bean texture was evaluated using the Ottawa texture measurement system for firmness (N mm⁻¹) and texture (N) (Voisey 1971). A visual examination for the degree of clumping of canned beans, the drain weight of washed-drained beans and the amount of water absorbed by a standard weight of beans were also used to judge cooking quality.

Performance

OAC Thunder matured approximately 2 d earlier than OAC Laser, and outyielded OAC Laser by 45 kg ha⁻¹ over 24 environments in 3 yr (Table 1). OAC Thunder is an indeter-

Table 1. Maturity, yield and seed weight of OAC Thunder and three checks in the Ontario Pulse Committee White Bean Cultivar Trials^z (1993–1995)

Cultivar	Yield (kg ha ⁻¹)				Maturity (d)				Seed weight (g 100 seed ⁻¹)
	1993	1994	1995	mean	1993	1994	1995	mean	
OAC Thunder	2950	3160	3089	3067	93.8	99.6	90.3	94.3	21.3
Centralia	2990	2506	2738	2745	93.8	97.0	90.2	93.4	22.3
OAC Cygnus	2740	2387	2799	2642	92.0	96.8	88.0	91.9	20.9
OAC Laser	3194	2924	2949	3022	96.0	103.8	92.3	97.0	19.9
SE ^y	93	99	102	59	1.4	1.3	0.9	0.8	0.2
No. of trials	6	8	10	24	6	8	10	24	24

^zTrials were conducted at Ailsa Craig, Brussels, Elora, Exeter, Kippen, St. Thomas, Shetland, Staffa, Winchester, and Woodstock, Ontario.^yBased on trial means of cultivars.**Table 2. Cooking quality and disease reaction of OAC Thunder and two checks in the Ontario Pulse Committee White Bean Cultivar Trials (1994–1995)**

Cultivar	Cooking quality						Disease reaction	
	Panel ^z (1–15)	Texture ^y		Packing ^x (1–5)	Drain wt. ^w (%)	Hyd. coef. ^v (%)	BCMV ^u	
		Plat.	Firm.				1	15
OAC Thunder	9.5	288	13.3	2.0	62.0	1.82	–	–
Centralia	8.5	345	12.9	2.2	61.3	1.86	–	–
OAC Gryphon	8.4	355	12.9	3.2	61.1	1.84	–	–
SE ^t	0.3	17	0.4	0.4	0.4	0.01		

^zPanel evaluation of cooked beans based on appearance, flavour and texture using a scale of 1–15: 1, poor; 15, excellent.^yTexture of washed-drained canned beans measured using wire extrusion cells, plateau force (N) and firmness (N mm⁻¹).^xDegree of clumping of canned beans as visually scored from 1 to 5: 1, no clumping; 5, over half of beans clumped.^wDrain weight: weight of washed-drained canned beans expressed as a percentage of unwashed-undrained weight.^vHydration coefficient: amount of water absorbed by 500 g of dry beans.^uResistance (–) or susceptibility (+) to bean common mosaic virus (BCMV) races 1 and 15.^tBased on six trial means of cultivars.

minate bush plant with erect stem and branches, and a vine similar in length to that found on OAC Gryphon. The cultivar can be successfully produced in either traditional wide (70 cm) rows or narrower (<40 cm) rows with direct combine harvesting.

Other Characteristics

OAC Thunder has a green hypocotyl in the seedling stage and white flowers which appear approximately 45 d after planting. Pods at maturity are light beige, although faint purple flecking may occur in some environments. Leaves are larger than those of OAC Speedvale, and similar to OAC Gryphon in both size and colour (grey-green). Seeds are dull white in colour and similar in size to OAC Cygnus. OAC Thunder has acceptable cooking quality, similar to Centralia (Table 2) and is resistant to races 1 and 15 of bean common mosaic virus (Table 2).

Maintenance and Distribution of Pedigreed Seed

Breeder seed of OAC Thunder is maintained by the Department of Plant Agriculture, University of Guelph, Guelph, ON, Canada N1G 2W1. Pedigreed seed is distributed through SeCan Association, 200-57 Auriga Drive, Nepean, ON, Canada K2E 8B2.

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